

Tumour–Immune Dynamics

Directed Reading Project(Spring 2026)

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Introduction • The Central Question

How can the qualitative geometry of a dynamical system — its equilibria, bifurcations, and invariant sets — explain and *predict* transitions between immune control and tumour escape?

Core thesis:

*Clinical transitions are bifurcation crossings in phase space,
not merely smooth parameter changes.*

Immunosurveillance

Stable $E_0 = (S/\mu_c, 0)$.
 $R_2 > 1$: any inoculum
cleared.

Dormancy

Stable coexistence
 (c^*, n^*) .
Can last years to decades.

Escape

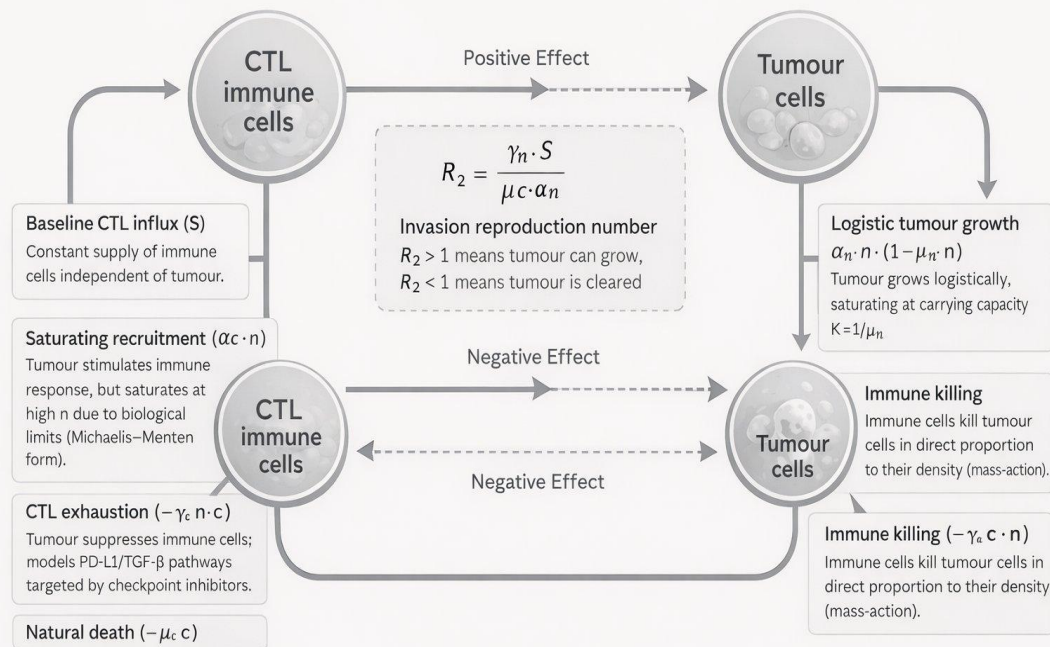
System crosses separatrix.
 $n \rightarrow K = 1/\mu_n$;
irreversible.

The ODE Model • The Two Equations

Understanding the Two Equations

$$dc = S + \frac{\alpha_c \cdot c \cdot n}{(\zeta_c + n) - \mu_c \cdot c}$$

$$dn = \alpha_n n \cdot (1 - \mu_n \cdot n) - \gamma_n c \cdot n$$



CTL immune cells $c(t)$:

$$\frac{dc}{dt} = \underbrace{S}_{\text{influx}} + \underbrace{\frac{\alpha_c c n}{\zeta_c + n}}_{\text{recruitment}} - \underbrace{\gamma_c n c}_{\text{exhaust.}} - \underbrace{\mu_c c}_{\text{death}}$$

Tumour cells $n(t)$:

$$\frac{dn}{dt} = \underbrace{\alpha_n n (1 - \mu_n n)}_{\text{logistic growth}} - \underbrace{\gamma_n c n}_{\text{immune killing}}$$

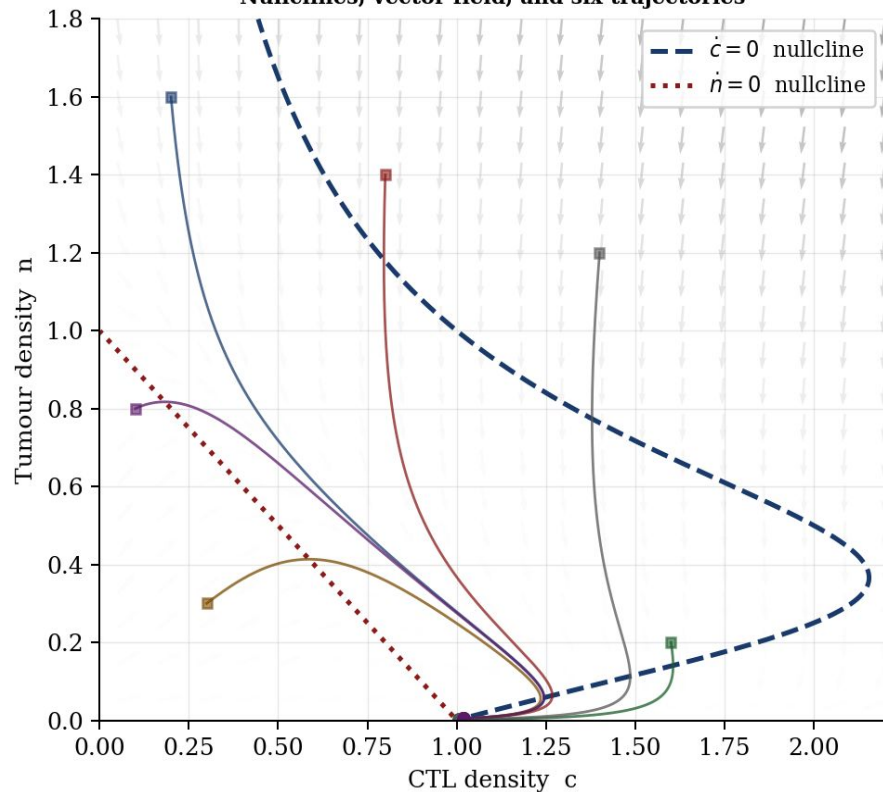
Invasion reproduction number:

$$R_2 = \frac{\gamma_n S}{\mu_c \alpha_n} \begin{cases} > 1 & \text{immune control} \\ < 1 & \text{tumour invades} \end{cases}$$

Checkpoint inhibitors $\downarrow \gamma_c \Rightarrow \uparrow R_2$.

Equilibria & Stability · Phase Portrait

Figure 2. Phase portrait (dormancy regime)
Nullclines, vector field, and six trajectories



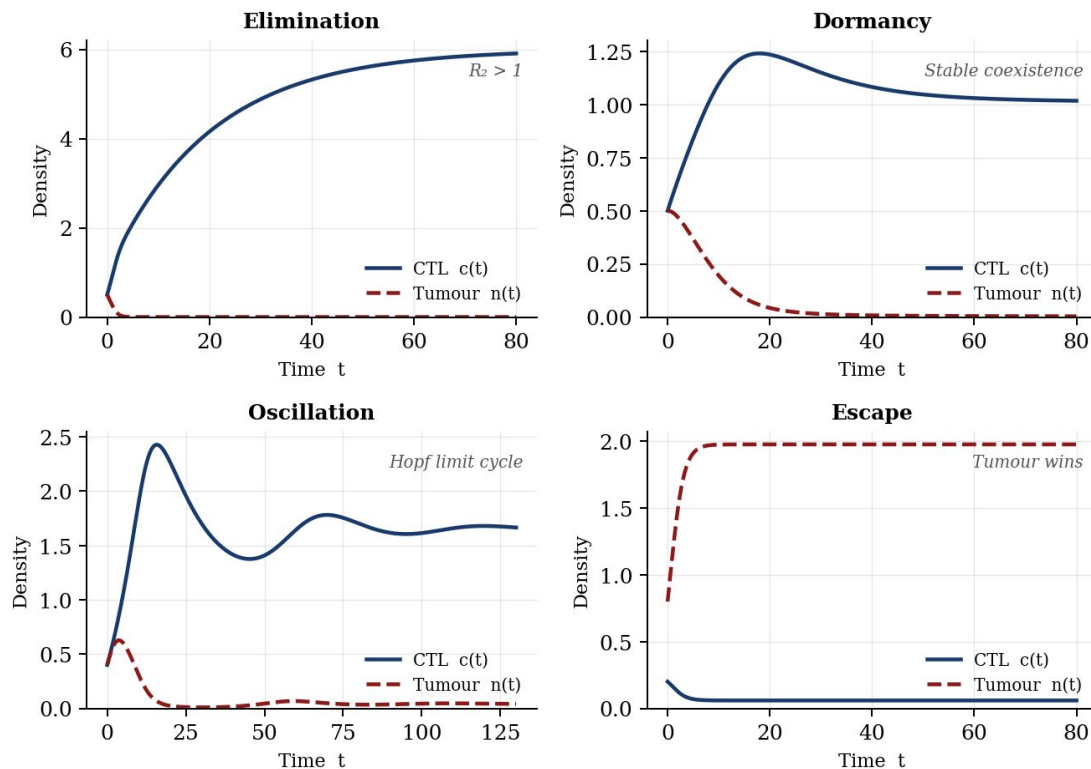
Jacobian at (c^*, n^*) :

$$J = \begin{pmatrix} \frac{\alpha_c n^*}{\zeta_c + n^*} - \gamma_c n^* - \mu_c & \frac{\alpha_c c^* \zeta_c}{(\zeta_c + n^*)^2} - \gamma_c c^* \\ -\gamma_n n^* & \alpha_n (1 - 2\mu_n n^*) - \gamma_n c^* \end{pmatrix}$$

Eigenvalue type	Classification
$\delta < 0$	Saddle — separatrix
$\tau < 0, \tau^2 > 4\delta$	Stable node
$\tau < 0, \tau^2 < 4\delta$	Stable spiral
$\tau > 0, \tau^2 < 4\delta$	Unstable spiral
$\tau = 0, \delta > 0$	Hopf point

Bifurcation Analysis · Four Regimes

Figure 1. Four dynamical regimes of the tumour-immune ODE system



Elimination

$R_2 > 1$, $n \rightarrow 0$.

Stable E_0 ; full immune control.

Dormancy

Stable (c^*, n^*) .

Subclinical; lasts years.

Oscillation

Hopf limit cycle.

Periodic relapse/regression.

Period $T \approx 2\pi/\sqrt{\delta}$.

Escape

$n \rightarrow K = 1/\mu_n$.

Separatrix crossed; irreversible.

Bifurcation Analysis • Sequence and Eigenvalue Locus

Figure 3. Bifurcation diagram — scanning γ_n

Dots: equilibrium. Vertical bars: oscillation amplitude. HC = homoclinic, TC = transcritical.

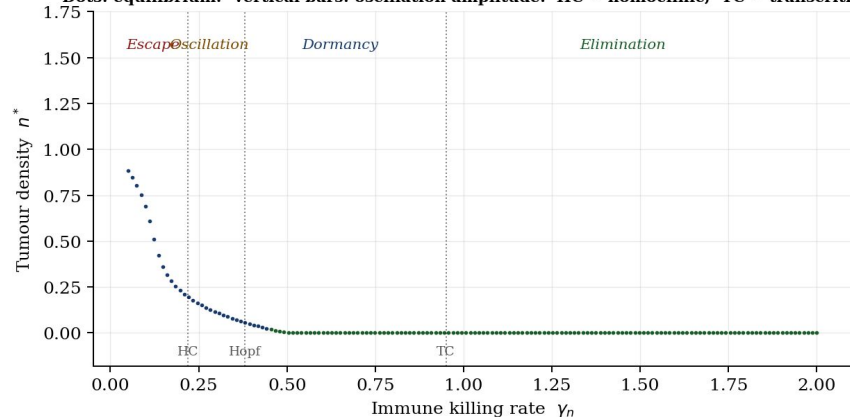
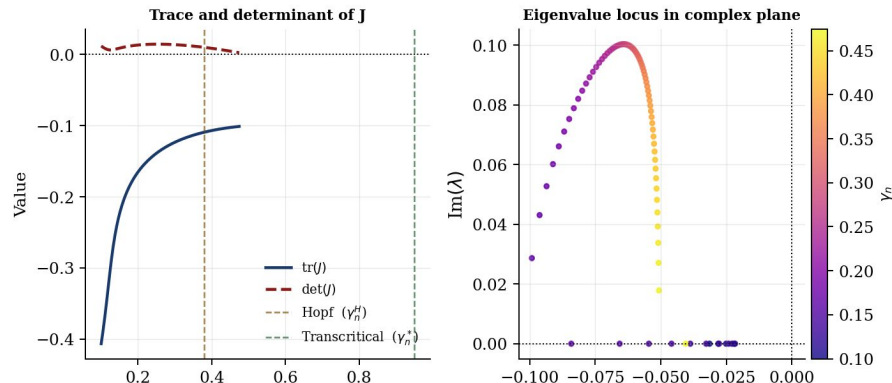


Figure 4. Stability analysis as γ_n varies



Eigenvalue locus in complex plane as γ_n varies. Colour = γ_n value (warm = low). Hopf crossing: $\text{Re}(\lambda)$ changes sign.

Four bifurcations as γ_n decreases:

1. **Transcritical**
 $R_2 = 1$
Invasion threshold

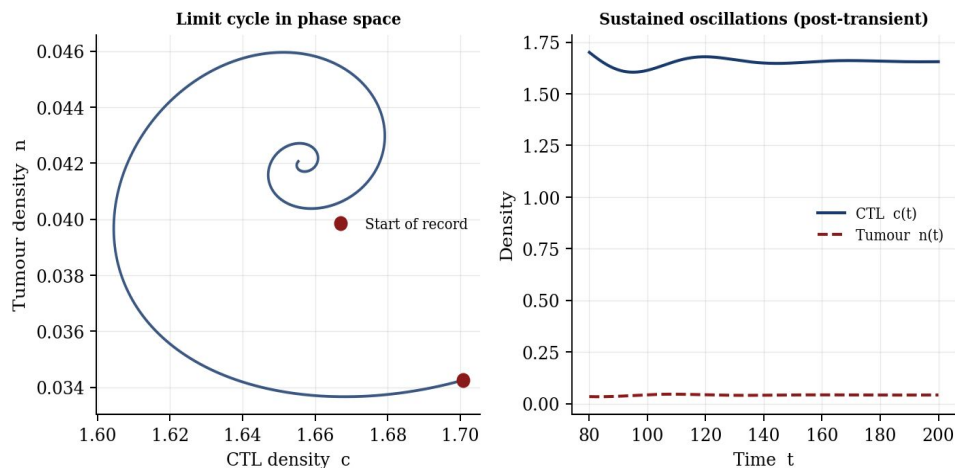
2. **Saddle-node**
Nullclines
tangent
Bistability, hysteresis

3. **Hopf**
 $\tau(J) = 0$
Limit cycle begins

4. **Homoclinic**
Cycle $\rightarrow$ saddle
Escape to carrying capacity

Bifurcation Analysis · Limit Cycle and Basin Structure

Figure 5. Hopf bifurcation: limit cycle dynamics



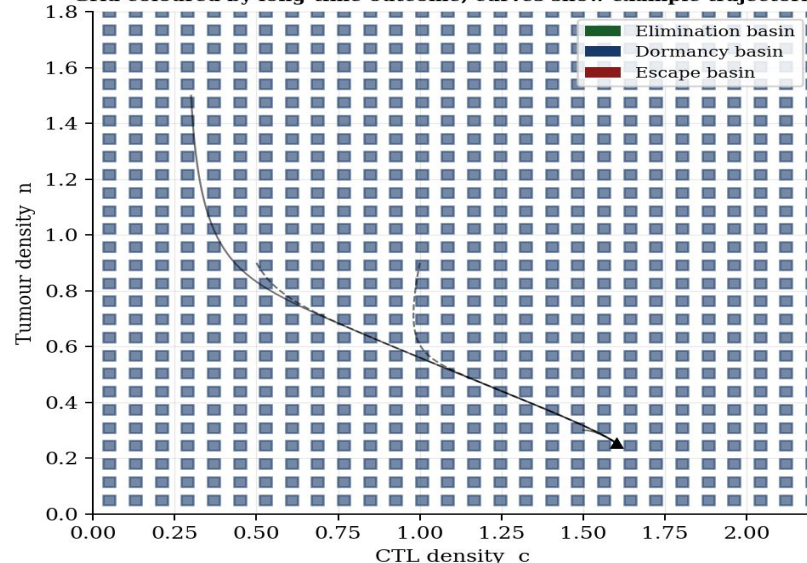
Post-transient limit cycle (Hopf regime). Left: phase orbit. Right: $c(t), n(t)$. Period consistent with $T \approx 2\pi/\sqrt{\delta}$.

Dulac criterion (multiplier $B = \frac{1}{cn}$): rules out limit cycles when $\text{tr}(J) < 0$

$$\frac{\partial(Bf_c)}{\partial c} + \frac{\partial(Bf_n)}{\partial n} = -\frac{S}{cn^2} - \frac{\alpha_c \zeta_c}{(\zeta_c + n)^2} - \mu_n < 0$$

When $\text{tr}(J) > 0$, the Poincaré–Bendixson theorem guarantees a stable limit cycle in a compact invariant region Ω .

Figure 6. Basins of attraction in the bistable regime
Grid coloured by long-time outcome; curves show example trajectories



Right: Basins of attraction in bistable regime. Outcome depends on initial (c_0, n_0) , not dose alone.

Model Extensions • PDE Immune Infiltration Wave

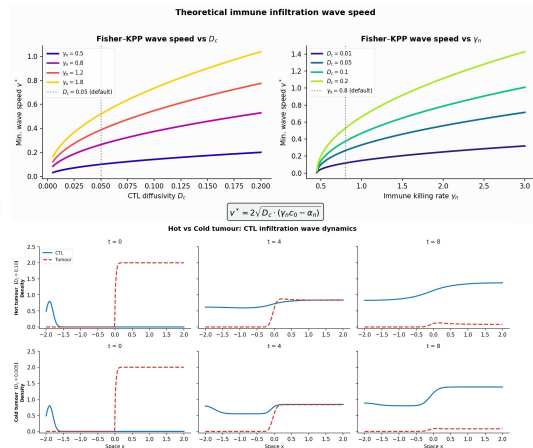
ODE limitation: ignores space. CTLs must physically infiltrate through stroma.

$$\frac{\partial c}{\partial t} = D_c \nabla^2 c + f_c, \quad \frac{\partial n}{\partial t} = D_n \nabla^2 n + f_n$$

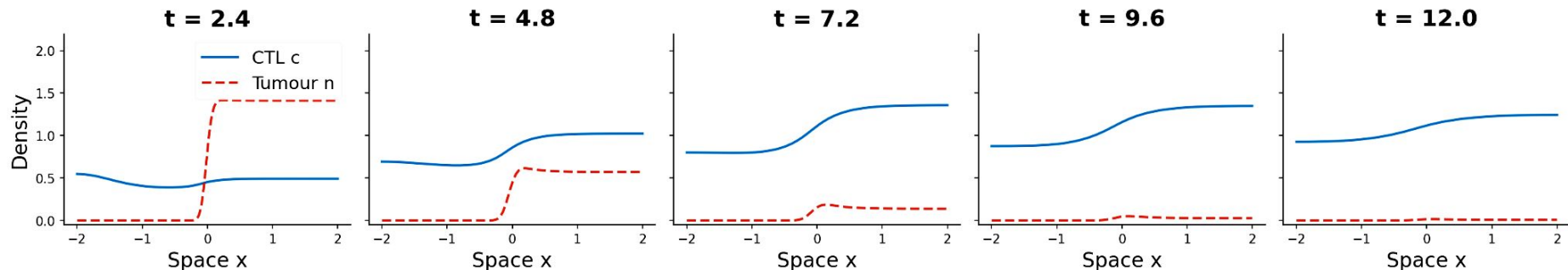
Fisher–KPP minimum wave speed:

$$v^* = 2\sqrt{D_c \cdot (\gamma_n c_0 - \alpha_n)}$$

- $D_c \gg D_n$ CTLs far more motile than tumour cells
- **Cold tumour** Dense stroma $\rightarrow$ low $D_c \rightarrow$ slow $v^* \rightarrow$ poor infiltration despite normal blood T-cell counts
- **Anti-stromal** Anti-FAP/VEGF raises $D_c \rightarrow$ faster wave $\rightarrow$ improved immunotherapy response
- **Keller–Segel** Adding $-\chi \nabla \cdot (c \nabla \phi)$ chemotaxis explains immune hot-spot patterning via Turing instability

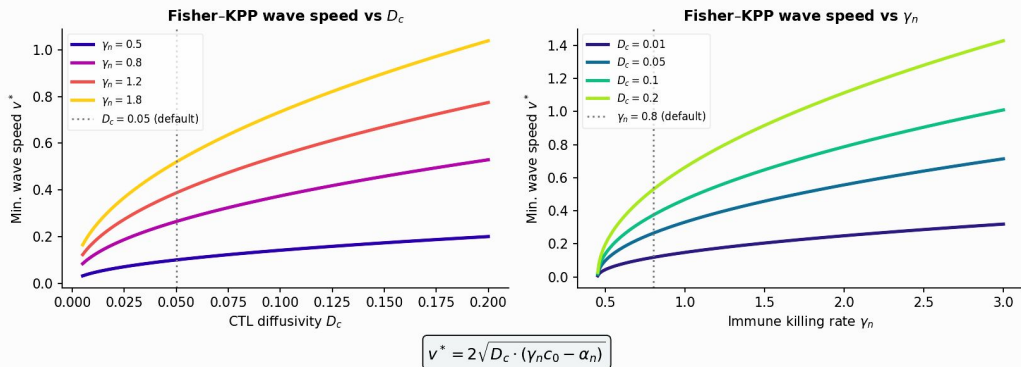


Reaction-diffusion PDE: immune wave infiltrating 1-D tumour
 $D_c = 0.05$, $D_n = 0.001$ (CTLs far more motile than tumour cells)



Model Extensions • PDE Immune Infiltration Wave

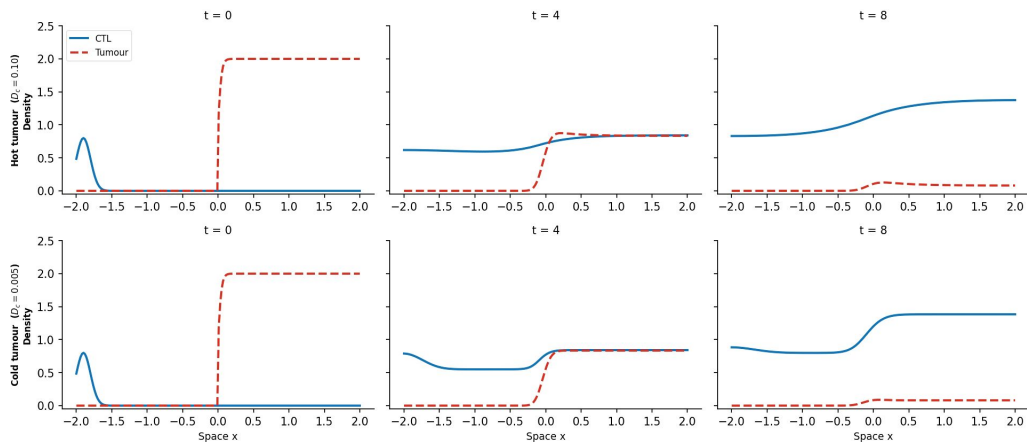
Theoretical immune infiltration wave speed



Fisher-KPP minimum wave speed:

$$v^* = 2\sqrt{D_c \cdot (\gamma_n c_0 - \alpha_n)}$$

Hot vs Cold tumour: CTL infiltration wave dynamics



higher D_c
 wave moves faster
 tumour cleared quickly

lower D_c
 wave moves slower
 tumour persists longer

Model Extensions · DDE, Stochastic & Optimal Control

Delay DDE

ODE ignores T-cell activation lag (3–14 days).

$$S + \frac{\frac{dc}{dt}}{\zeta_c + n(t-\tau)} = \alpha_c c(t-\tau) n(t-\tau) - \gamma_c n c - \mu_c c$$

Characteristic eq.:

$$\det(J_0 + J_1 e^{-\lambda\tau} - \lambda I) = 0$$

Critical delay $\tau_H \approx 5\text{--}8\text{ d}$ creates a Hopf bifurcation *independent* of γ_n . ODE overestimates dormancy stability.

Stochastic Langevin

ODE ignores fluctuation-driven escape.

$$dc = f_c dt + \sqrt{c/\Omega} dW_c$$

Kramers mean first-passage time:

$$T_{\text{esc}} \approx \frac{2\pi}{\sqrt{|\lambda_s^{(1)}| |\lambda_s^{(2)}|}} e^{\Omega \Delta U}$$

10–15 % drop in $\Delta U \Rightarrow T_{\text{esc}}$ falls by orders of magnitude.

Perioperative window: surgery transiently $\downarrow \Delta U$; basis for beta-blockade + COX-2 inhibition timing.

Optimal Control

ODE gives no therapy design.

Cost functional:

$$J = \int_0^T [n + w_1 u_1^2 + w_2 u_2^2] dt + Bn(T)$$

Hamiltonian:

$$H = n + w_1 u_1^2 + w_2 u_2^2 + p_1 f_c + p_2 f_n$$

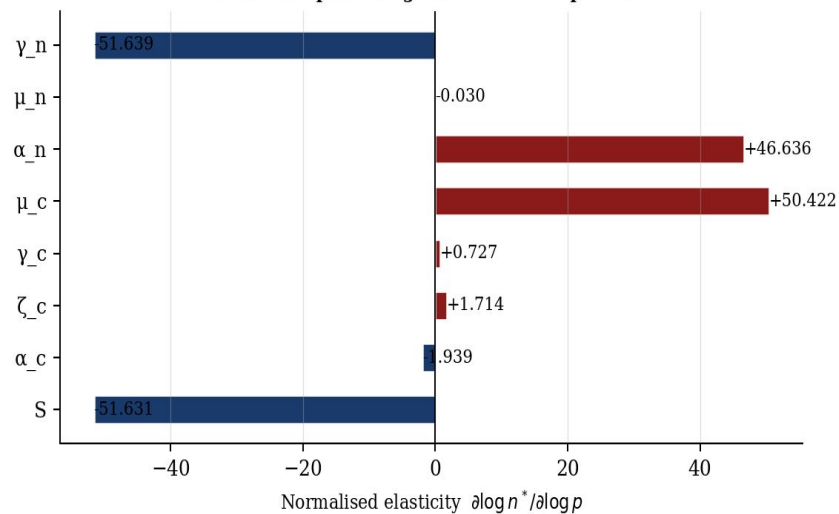
Pontryagin $\Rightarrow$ **bang-singular-bang**:

1. **Bang:** cross separatrix (high dose)
2. **Singular:** hold basin at min. cost
3. **Off:** immunity self-sustaining

Justifies induction–maintenance protocols.

Clinical Applications · Sensitivity & Clinical Targets

Figure 7. Local sensitivity of dormancy equilibrium n^* to each parameter
Red: tumour-promoting. Blue: immune-protective.



Normalised elasticity: $\frac{\partial \ln n^*}{\partial \ln p_i}$

Red: tumour-promoting Blue: protective

Clinical scenario	Math mechanism	Intervention
Checkpoint inhibition	$\downarrow \gamma_c \Rightarrow R_2 > 1$	Anti-PD-1 / PD-L1
Acquired resistance	Re-cross saddle-node	Combination therapy
CAR-T minimum dose	$c_{th} > c_{sep}(n_{pt})$	Lymphodepletion
Perioperative relapse	Kramers: $\Delta U \downarrow$	Beta-block, COX-2
Combination therapy	$R_2 = \frac{\gamma_n S}{\mu_c \alpha_n}$	Multiplicative gain

CAR-T separatrix criterion:

$$c_{therapy} > c_{sep}(n_{patient})$$

Conclusion • Summary

Four dynamical regimes

Regime	Structure	Bifurcation
Elimination	E_0	Transcritical
Dormancy	(c^*, n^*)	Saddle-node
Oscillation	Limit cycle	Hopf
Escape	$n \rightarrow K$	Homoclinic

Six extensions

Extension	Insight
Reaction–diffusion	Wave speed
Chemotaxis	Hot/cold spots
Free boundary	Tumour geometry
Delay	Oscillations
Stochastic	Escape time
Control	Therapy design

Central message

Nonlinearity is not a complication — it creates structure. Each regime is a *distinct dynamical state*, separated by bifurcation thresholds.

Thesis

Therapy should move the system to a different dynamical basin — not just reduce tumour size.

Parameter Reference

Symbol	Meaning	Typical range	Therapeutic lever
S	Baseline CTL influx	0.05–0.30	IL-2, vaccines
α_c	CTL recruitment strength	0.10–0.80	Neoantigen load
ζ_c	Saturation half-constant	0.20–1.00	—
γ_c	CTL exhaustion rate	0.05–0.50	Anti-PD-1 (↓)
μ_c	CTL natural death	0.02–0.20	IL-7, IL-15
α_n	Tumour intrinsic growth	0.10–1.00	CDK4/6-i
μ_n	$1/K$ (carrying capacity)	0.50–2.00	Anti-VEGF
γ_n	Immune killing rate	0.05–2.00	CAR-T (↑)

Thank You

Questions?

S
αc
γc
μc

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